

Bank Customer Churn Analysis

A Business Analyst Case Study — Python / Pandas EDA

1. Project Overview

This project analyzes customer churn for a retail bank operating across France, Germany, and Spain. The goal was to identify which customer segments are most likely to leave, quantify the scale of the problem, and surface actionable, data-backed recommendations — using Python and pandas for the full workflow instead of SQL/Power BI, to demonstrate range across the toolset used in the other two portfolio projects.

Dataset: Kaggle-style "Bank Customer Churn" dataset, provided as a deliberately messy two-sheet Excel workbook (Customer_Info and Account_Info), requiring merging, validation, and cleaning before analysis.

Tools: Python, pandas, matplotlib, Jupyter (VS Code).

2. Data Cleaning

The raw workbook contained several realistic data-quality issues that had to be resolved before any analysis could be trusted:

- Merged two sheets (Customer_Info, Account_Info) on CustomerId; verified row counts and checked for unmatched IDs after an outer join.
- Removed 4 exact duplicate customer records.
- Stripped currency symbols (€) and converted Balance and EstimatedSalary to numeric; identified and nulled out a -€999,999 sentinel value used as a hidden "missing" marker in EstimatedSalary.
- Discovered HasCrCard values were incorrect in the working file (identical distribution to IsActiveMember, a red flag) — cross-checked and corrected all values using a clean reference extract, joined on CustomerId.
- Verified IsActiveMember was accurate despite an initial suspicious match with the (still-broken) HasCrCard column — confirmed via a zero-mismatch check rather than assuming.
- Standardized inconsistent Geography labels: "France", "French", and "FRA" unified into a single "France" category.
- Encoded Gender, HasCrCard, and IsActiveMember as binary numeric fields.
- Resolved a duplicate Tenure_cust / Tenure_acc pair (values matched 100%) into a single Tenure column.
- Filled the small number of remaining missing values: median for Age and EstimatedSalary, "Unknown" for Surname (kept, rather than dropped, since it supports naming specific at-risk customers in stakeholder-facing output).

Result: a fully validated 10,000-row, 13-column dataset ready for analysis, with every cleaning decision checked against evidence rather than assumed.

3. Key Findings

Baseline churn rate: 20.4% of customers (2,037 of 10,000) have churned — roughly 1 in 5.

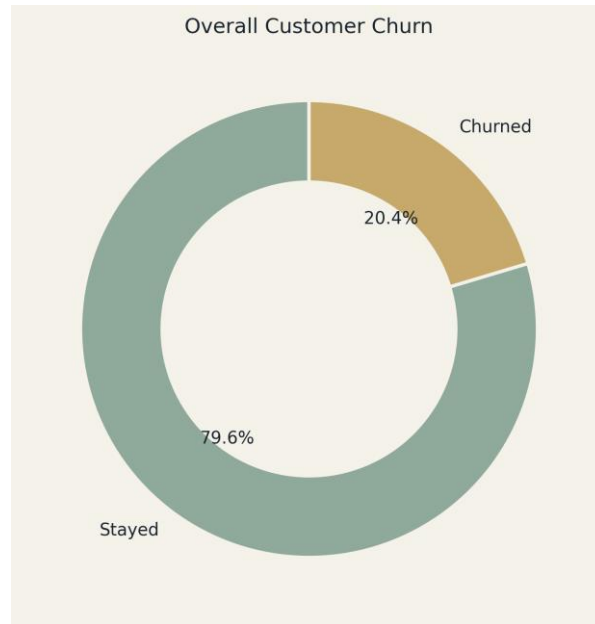


Figure 1 — Overall churn split.

3.1 Geography

Germany churns at nearly twice the rate of France and Spain — a clear, actionable regional signal.

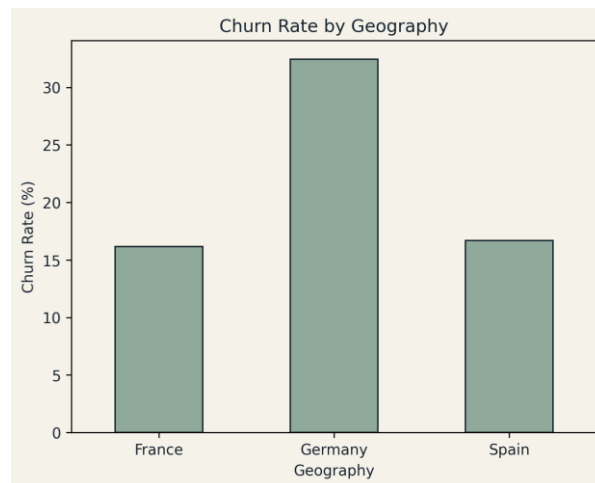


Figure 2 — Churn rate by country.

Germany	32.4% churn
France	16.2% churn
Spain	16.7% churn

3.2 Number of Products

Customers holding 2 products are the most loyal segment (7.6% churn). Customers with 3 or 4 products show very high churn (82.7% and 100%), though these groups are small (266 and 60 customers respectively) and should be treated as a flag for further investigation rather than a fully proven driver.

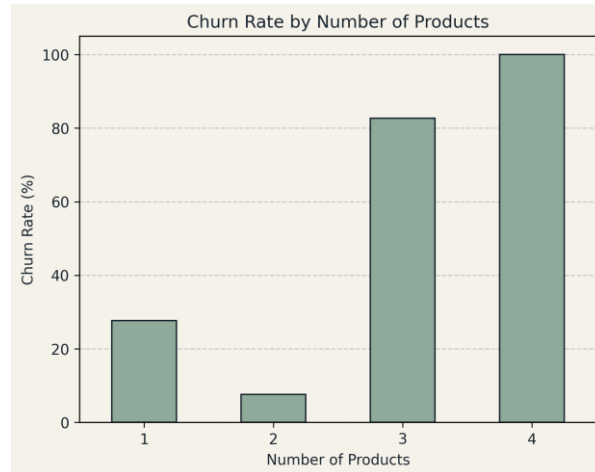


Figure 3 — Churn rate by number of products held.

3.3 Membership Activity

Inactive members churn at almost double the rate of active members (26.9% vs 14.3%), on a well-balanced sample — one of the most reliable and actionable findings in the analysis.

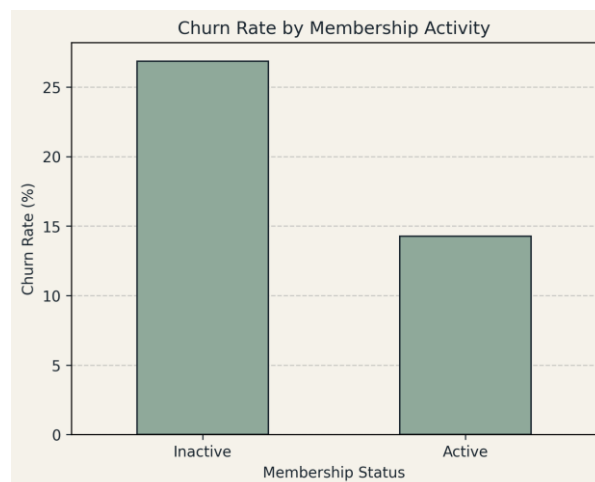


Figure 4 — Churn rate by membership activity status.

3.4 Gender

Female customers churn more than male customers (25.1% vs 16.5%).

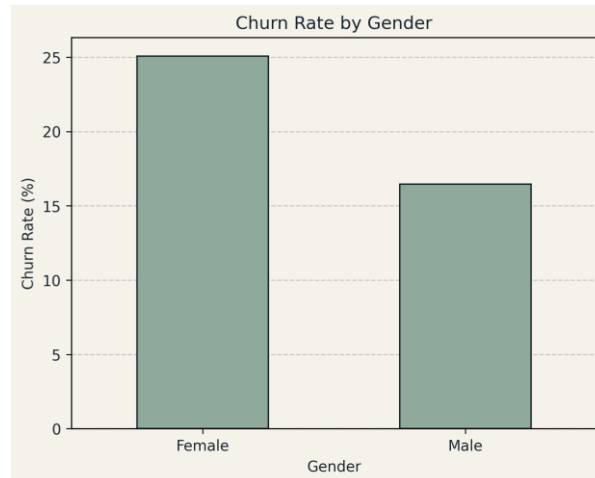


Figure 5 — Churn rate by gender.

3.5 Age

Churn is not linear with age. It rises through the 40s, peaks sharply in the 51-60 age band (over 56% churn), then drops again for 61-70 and 70+. This is more precise and more useful than a simple average comparison — it identifies a specific pre/early-retirement risk window rather than "older customers churn more" in general.

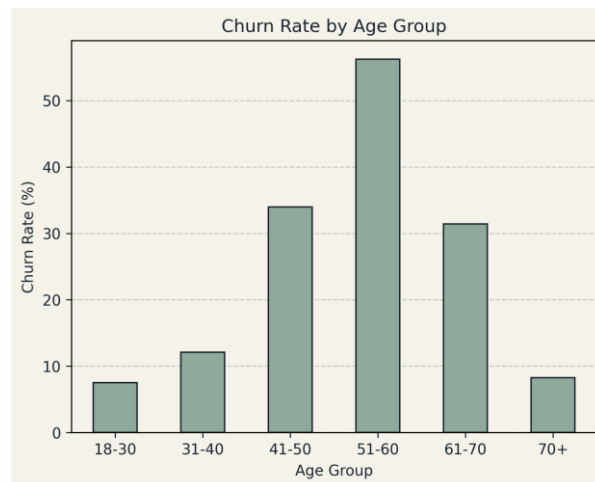


Figure 6 — Churn rate by age group.

3.6 Balance

Customers with a €0 balance churn the least (13.8%, on a large, reliable sample of 3,617 customers). Mid-range balances hover between 20-35% churn. The 200k+ segment shows the highest churn rate (~56%) but is based on only 34 customers, so it is reported as directional, not conclusive.

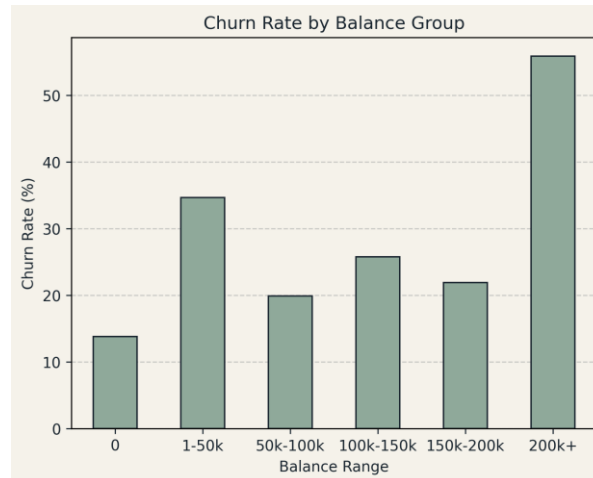


Figure 7 — Churn rate by balance range.

3.7 Age & Balance Distribution (Box Plots)

Box plots confirm the pattern from the segment analysis: churned customers have a higher median age and a higher median balance than retained customers, with the churned group's age distribution shifted noticeably upward.

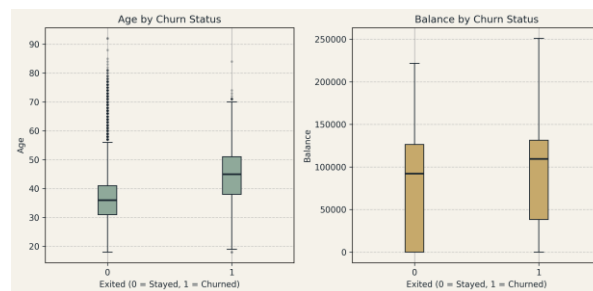


Figure 8 — Age and Balance distribution by churn status.

3.8 Correlation Overview

A correlation heatmap across all numeric fields confirms the segment-level findings: Age (0.29), IsActiveMember (-0.16), and Balance (0.12) show the strongest linear relationships with churn. CreditScore, Tenure, and HasCrCard show negligible correlation, confirming they are not meaningful churn drivers. Note that these correlations are individually modest — consistent with churn being driven by a combination of factors rather than any single strong variable, and with the fact that correlation captures linear relationships only, missing non-linear patterns like the age-51-60 spike found in the segment analysis.

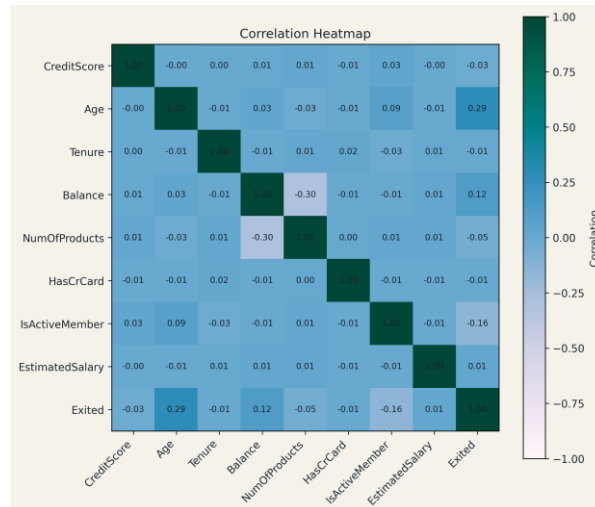


Figure 9 — Correlation heatmap across numeric variables.

4. Factors Checked with No Meaningful Pattern

Equally important to a rigorous analysis is reporting what was tested and ruled out, rather than only reporting positive findings:

- CreditScore — minimal difference between churned (645.4) and retained (651.9) customers.
- Tenure — churn rate stays flat (~17-23%) regardless of how long someone has been a customer.
- HasCrCard — negligible difference (20.8% vs 20.2%) once the column was corrected.

5. Recommendations

- Investigate Germany specifically — regional service quality, pricing, or competitive pressure likely explains the 2x churn gap versus France/Spain.
- Prioritize re-engagement campaigns for inactive members — this is the largest, most reliable, and most actionable lever identified.
- Build a targeted retention track for the 51-60 age segment, rather than treating "older customers" as one uniform risk group.
- Review the onboarding/relationship-deepening path for customers with only 1 product, and audit the 3-4 product segment for a possible data or process issue given the unusually high (and small-sample) churn rate.
- Treat the €200k+ balance finding as a hypothesis to validate with more data, not a confirmed driver, given the small sample size.

6. Next Steps

A predictive model (e.g. logistic regression or random forest) is planned as a follow-up phase, to move from descriptive findings ('who churns and why') to a predictive score ('which specific customers are at risk right now'), and will be published as an updated version of this project.